Is what we see a projection? Objects as the common part of two points of view, tested on a quantum-Darwinism model

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Abstract

When two observers correctly read the same quantity and compare their records, they find them in agreement, within the measurement errors. This deposit proposes to read the object both of them see as the common part of their records : what each reads from its own record, without looking at the other's, with the same value as the other (the Gács–Körner common part, exact or with a very small tolerance when the copies are imperfect). The key word is projection: the reading projects a record onto a simpler question, like a shadow, and the projections of two records coincide, in the model within a chosen tolerance. In the author's framework the observed object is a transposition and non-local projection of the observer's "observation matrix"; in the model of this work the matrix is represented, by a choice of the AI, by the raw record of what the observer collected. What two projections share is the trace of one non-local relation, non-local in the sense of not located in space and time, not of an influence at a distance.

Ugo Lanciano, October 2026. Licence CC BY 4.0 (texts); code MIT. The same text in Italian is in LEGGIMI_IT.md.

Version 2.62.1 (25 October 2026): corrections and a new section. An independent check requested by the author found places where version 2.62 presented as a result about the thesis what depended on translation choices made by the AI and not declared. This version corrects them and adds section 9, on an object with more values; the programs, the calculations and the numbers of 2.62 do not change.

1. The author's criterion C1 concerns compared records only. Version 2.62 had added a second part (a very likely agreement, even without comparison, would require a common object), which was not in the criterion. Without that part, "the same matter" (T-1) is not excluded but does not conform to the criteria, and "the exact common part" (T-4s) is compatible.
2. The author's "matrix" had been rendered by the raw photon record of the model: the numbers on the small common part of the whole record (8.5 %, 3.4 %) concern that choice, not the author's matrix.
3. The common divisor of the two GCDs (T-3) is compatible as an arithmetic rule of the common part: it is computed from the joint distribution alone (test P1, already registered); as for T-4, the criterion on distances (C5) is met only together with triangulation, which is standard geometry.
4. Identifying the "nexus" with the source variable was a choice of the AI: copies of the same fact (T-2) and chains of copies (T-5) cannot be judged in the model.
5. The paradoxical side of the thesis concerns the non-local link, not the recorded common part.
6. The yes-or-no of the common part holds for the 1-bit object of the model; with more values, within the class of readings tested, the common part grows in steps (section 9).
7. Distance from luminosity (T-7L) was not "non-reciprocal": it needs the total number of copies emitted, a datum distinct from the mass, which in the model is not derived from it; triangulation is standard geometry and does not translate the author's "fractionation".
8. The probabilistic definitions tested use the distributions of quantum mechanics: by construction, in the cases examined, they give no different statistics.
9. The separation from mass holds in the static model.
10. New section 9, «An object with more values»: within the class of readings tested the common part grows in steps, with locally blurred copies coarse fractions first, but the steps do not follow the divisors of the grid; the common divisor of the two GCDs is the ceiling or a stage, depending on where the noise enters.

Details: 04_controllo_262_264/PIANO_CORREZIONI_v1.md (in Italian, as registered); the work on the object with more values is in the folder 05_oggetto_piu_valori.

In brief

When two observers correctly read the same quantity and compare their records, they find them in agreement, within the measurement errors: no case of systematic disagreement is known. We propose to read the object that both of them see as **the common part of their records**: what each of them reads from its own record, without looking at the other's, with the same value as the other. Eight mathematical formulations of this idea were written down and registered before any calculation, and tested on a small model of the kind used in quantum Darwinism. The formulations that survive measure the common part exactly, with a tolerance or in a graded way, together with that of the common divisor of the two GCDs; "the same pattern" fails. In the model, with imperfect copies, the exact common part is zero until the records are compared, and it becomes the whole object when B faithfully copies A's reading. In the model, where the object is worth 1 bit, the common part with tolerance behaves like a yes-or-no: it is zero below a threshold number of copies read, and it is essentially the whole object above it. The probabilistic definitions tested use the probabilities of quantum mechanics: in the cases examined, by construction, they give no different statistical prediction. The key word is **projection**: each observer sees the object through a projection of its own record, like a shadow, and the projections of the two records coincide (in the model, within the chosen tolerance). In the author's framework, what two projections have in common is the trace of one and the same non-local relation, that is, a relation not located in space and time.

An object with more values (§9, new in this version). With an object that can be in 12 positions instead of two, read by blurred copies, the common part computed within the class of readings tested no longer arrives all at once: it grows in steps, first the halves, then the thirds, the quarters and so on, one boundary at a time, within a few tens of copies. The steps, however, do not follow the divisors of the grid, and the common divisor of the two rulers becomes the ceiling of the common part or only a stage, depending on where the noise enters.

How to read this text. Section 2 explains the thesis without formulas and should be read first; Section 2.2 explains the key word, projection, with three images. Section 1 is a review of existing studies, useful above all to readers who know the field: on a first reading it can be skimmed. Section 3 describes the model on which the thesis was tested; Section 4 gives the results, starting with three examples that can be done by hand. Section 9, new in this version, deals with an object with more values and can be read on its own. Technical words are explained in brackets the first time they appear, and again in the glossary at the end.

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1. What is already known

Compared records agree. In every experiment in which two observers have correctly read the same quantity and compared their records, they have found them in agreement, within the measurement errors. This holds in Wigner's-friend experiments too (a scenario in which one observer, the friend, makes a measurement shut inside an isolated laboratory, while a second observer outside, Wigner or the "superobserver", treats the whole laboratory as a single quantum system): in the description of these experiments the superobserver who opens the laboratory and asks the friend what was seen finds the outcome the friend recorded (Bong et al. 2020).

Objectivity as redundancy: quantum Darwinism. For Zurek and collaborators a fact is objective when many copies of the same information are spread through the environment, so that many observers can read it, each from a different fragment, without knowing it beforehand and without disturbing the system (Zurek 2003, 2009; Ollivier, Poulin and Zurek 2004). The numbers are huge: a one-micron dust grain in sunlight imprints its position on the scattered light about a hundred million times in a microsecond (Riedel and Zurek 2010). The number of copies needed to read a fact is about $\ln(1/\delta)$ ($\ln$ is the natural logarithm) divided by the quantum Chernoff information of one copy (a measure of how much a single copy helps to tell the two possible values apart), where δ is the error one accepts (Zwolak, Riedel and Zurek 2014: an estimate for many copies); in the ideal case of light arriving equally from all directions, the readable copies disappear (Riedel and Zurek 2011). The precise state structure that guarantees objectivity is the spectrum broadcast structure (the state in which every fragment of the environment, on its own, carries the value of the system) (Korbicz, Horodecki and Horodecki 2014; Horodecki, Korbicz and Horodecki 2015; Le and Olaya-Castro 2019; review by Korbicz 2021); the objectivity of observables emerges generically, that of outcomes does not: what can be measured becomes the same for everyone almost always, whereas what is found does so only with suitable interactions (Brandão, Piani and Horodecki 2015). Laboratory measurements exist with photons (Ciampini et al. 2018; Chen et al. 2019) and with a nitrogen-vacancy centre in diamond (a defect in diamond that behaves as a small quantum system) (Unden et al. 2019).

Existence as a matter of degree. Zurek speaks of "relatively objective existence": the existence of a state is not an intrinsic attribute but arises from redundant copies, and it is a matter of degree (Zurek 2022). This is the closest precedent to what we propose.

Relative facts. In relational quantum mechanics facts are relative to systems (Rovelli 1996); they become "stable" for another system through decoherence (the disappearance of interference in the behaviour of the system alone when it

interacts with its environment; coherence is not destroyed, it passes into the correlations with the environment) (Di Biagio and Rovelli 2021); for different observers to agree when they compare, an extra postulate is needed, the cross-perspective links: if B measures the variable in which A recorded an outcome, B finds A's outcome (Adlam and Rovelli 2022). The theory faces known objections (Pienaar 2021; Brukner 2021, "qubits are not observers").

Wigner's-friend theorems. Frauchiger and Renner (2018), Brukner (2018) and Bong et al. (2020) show that certain natural assumptions cannot all hold together; among them, in different forms, that what each observer has recorded holds in the same way for everyone (in Bong et al.: the Absoluteness of Observed Events). Photonic experiments test them: Proietti et al. (2019) measure $S = 2.416 \pm 0.075$ (S is a combination of correlations that could not exceed 2 if all the assumptions held): the bound is exceeded by more than five standard deviations, that is, well beyond the measurement error; Bong et al. (2020) violate a Local Friendliness inequality (a bound that the measured correlations should respect if all those assumptions held together). Cavalcanti and Wiseman (2021) give the taxonomy; Wiseman, Cavalcanti and Rieffel (2023) propose a friend that is an artificial intelligence running on a quantum computer. Rivlin, Engineer and Baumann (2025) show that when partial copies of the friend's outcome leave the laboratory, the room for non-absolute events shrinks: their CHSH-type inequality becomes $\leq 2 + 4\varepsilon$, where $1 - \varepsilon$ is the probability of recovering the friend's outcome, and for $\varepsilon \geq 0.207$ there is no violation. These theorems concern events that are **not** compared.

Agreement as a principle. Two agents with the same prior, whose estimates are common knowledge (each knows the other's estimate, knows that the other knows theirs, and so on), cannot "agree to disagree" (Aumann 1976). The same holds with quantum correlations; disagreement becomes possible only with correlations stronger than quantum ones (Contreras-Tejada et al. 2021). Leifer and Duarte (2026) show that the theorem holds in any theory with the law of total probability. Cepollaro and Di Biagio (2026) show that no state of the world is needed: a joint distribution of the records (a single table of the probabilities of all combinations of records) suffices, and the theorem can fail only where such a distribution is missing, that is, in Wigner's-friend scenarios.

The common part of two variables. Gács and Körner (1973) define the common part of two random variables: the richest quantity that can be computed from either one with the same result.

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2. The thesis, motivated and described

2.1 What it says

The thesis concerns **observed physical objects**. For an observer, an object is not something that already exists in a given space and is then seen: it is a part of the observer's **observation matrix**, as the author calls it (in the AI's reading of the author's framework, the structure of counting, multiples and fractions, seen from the observer's point of view; in the model of this work it is represented, by a choice of the AI, by the record of what the observer has collected), transposed and projected into smaller parts. "Transposed" means that this part of the matrix presents itself as something located elsewhere, at a distance from the observer; "projected", that only an outline of it is kept (§2.2). Two observers see "the same object" because their matrices, the author says, while not equal as a whole, are equal in large percentage. In the model, what the two records have in common is their reading of the object (about 1 bit, with a small tolerance), and the common object **is** that common part, not something that lies underneath the records and produces them. On the whole raw record of the model, a choice of the AI, the common part is small (§4): the model does not test the author's sentence on the matrix.

Three words, defined at once:

- **observer:** any physical system that records and compares; not a mind. This is the operational definition used in Wigner's-friend experiments (Proietti et al. 2019: an observer is a system that extracts information and stores it in a physical memory);
- **record:** everything an observer has recorded, as a physical state of its memory;
- **reading (or projection):** what the observer extracts from the record by bringing it down to a simpler question; it is a function of the record and usually discards part of it (§2.2).

Two things not to confuse. The chair, too, is a part of the structure, with a state and local relations of its own (with the air, with the light that hits it): in this sense it is an autonomous entity, and the thesis does not deny it. The thesis says something else: what of the chair exists *for* A is the common part between the state of the chair and A's record (with imperfect copies, the common part with a very small tolerance: §4); what A and B agree on is the common part between their two records. When the two observers have collected enough copies, these three common parts (chair and A, chair and B, A and B) are the same quantity, and this is what "they see the same chair" means. In the model of §3, with 23 copies that are wrong 10 % of the time, A's reading differs from the state of the chair with probability 4.7×10^{-7}, and B's reading differs from A's with probability 9.4×10^{-7}. No part of the structure is privileged: the chair, A and B are three

parts, and the relation between any two of them is measured in the same way.

2.2 Projection: the shadow, the ruler and the non-local phase

The key word of the thesis is **projection**. Three images make it visible.

The shadow. A lit body casts a shadow on the wall. The shadow keeps the outline and loses everything else: the colour, the inside, the depth. Different bodies can cast the same shadow. A projection always does this: it takes something rich onto something poorer, and loses a great deal on the way.

The record is very rich; the reading is its shadow. The record of someone looking at a chair contains a great many details that belong to that observer alone: which photons arrived and when, the noise of the instrument, the point of view. The record of a second observer contains just as many, all different. Compared as wholes, the two records hardly resemble each other. To see the chair, each brings their own record down to a simpler question, "where is the chair?", and discards the rest, as the shadow discards the inside of the body. It is in this projection, and only there, that the two records coincide, in the model within the chosen tolerance: the two shadows have the same outline. The common object is what appears in the two projections with the same value; mathematically, the Gács–Körner common part is exactly this, the richest projection two records have in common; with imperfect copies one uses its form with a very small tolerance (§4), because the exact one is zero. The numbers of §4 show it: with 23 copies read, the common part is about 1 bit, that is, almost the whole projection (the question "where is the chair?" is worth 1 bit, and the record carries 99.9998 % of it), but only 8.5 % of the whole record, which by then holds about 11.8 bits; with more copies the whole record grows and its common part shrinks (3.4 % at 61 copies), while the projection stays almost entirely common. If one asked for the whole records to coincide, two observers would never see the same object. These numbers concern the raw record of the model, in which every copy also carries its own noise: this is a choice of the AI, not the author's matrix, which for two nearby observers the author calls "equal in large percentage"; the model does not test that sentence.

The ruler: a projection in fractions. Reading a position on a ruler is a projection: the exact position is brought to the division it falls in, that is, to a fraction of the unit, and the rest is lost. Each observer has its own ruler, with its own divisions: one in sixths, one in ninths. Their readings of the same position differ, but they have in common the reading on the divisions the two rulers share, the thirds. This is the sense in which the projection is "in fractions": each observer reads the world in the fractions of its own scale, and what two observers have in common lives in the common fractions.

What casts the shadow: the non-local phase. In the usual view the shadow is cast by a body sitting in a space given in advance. The author's framework puts something else underneath: a **non-local phase**, made of relations without space and without time. "Non-local" here does not mean an influence at a distance, nor one faster than light: it means that the relation is not in a place, and is present in the same way in every place and at every instant. The local world is the same reality counted from a point: the copies of a relation, which whoever reads them arranges in a row (time) and side by side (space). In a reading by the AI (not demonstrated: identifying the copies of the framework with scattered photons is a choice), every photon scattered by the chair is a copy of the same relation, and between two copies of the same thing there is no distance: in this reading, this is why two people see the same chair with different photons. Their records are two counts of one and the same non-local relation, made from two different points, and the common part is the trace of that relation in the two counts. This is how the framework reads why the shadows coincide: underneath, the relation is one. In the author's framework, the observed object thus arises as a **non-local transposition and projection of the observation matrix**: when the observer projects its own matrix and discards the private part, what it isolates is not a datum locked inside its memory, but the trace of the non-local relation itself, which has no distance between its copies. This is how the framework reads why the projection of A's matrix coincides with the state of the chair and with the projection of B's matrix (in the model, up to probabilities of the order of one in a million: §2.1). It is the same idea with which the framework reads entanglement: the two distant measurements exchange nothing, they are the two ends of a single relation. And this is also how it reads the hardest side of the thesis: what has no non-local link with us casts no shadow in our record, and for us it does not exist.

What is lost. What the projection discards is not a defect of the reading: it is the part of the record that belongs only to whoever holds it, its point of view. Two different records can give the same projection, as different bodies cast the same shadow: this is why the common object is not everything in the records, but what remains the same after the projection.

2.3 Why think of it this way

1. **No observer reaches an object except through its own record.** This is an operational fact, not a philosophy: everything an observer knows about an object is written in some physical state of its own.
2. **Physics has already gone part of the way.** In relativity the simultaneity of distant events belongs to the observer; what everyone shares are coincidences and causal order. In quantum mechanics the state is relative (Rovelli), the outcomes of a sealed laboratory are not absolute (Brukner; Bong et al.) and objectivity arises from copies (Zurek). One step remains: to say that the observed object itself, and not only its state or its properties, is

what the records have in common.

3. **The author's framework asks for it.** The framework from which this work arises starts from a postulate: an entity exists for another to the extent of what they have in common, and at the same time the other exists for it. If existence is a relation, so is the observed object: what two observers have in common is what exists for both of them. The framework also distinguishes a non-local phase, of relations without space and without time, and a local phase, which is the same reality counted from a point (§2.2). The author's own words are in the background file, kept separate from this text.

2.4 How it reads some known facts

- **Why compared records never disagree.** In the ideal case, comparing means that B makes a faithful copy of A's reading. From then on that reading is in the common part, and on the common part two observers cannot disagree: that is what "common" means. The cross-perspective links of Adlam and Rovelli are read as this ideal case, and the size of the common part says how much the two shared even before the comparison. This is not a proof of their postulate: it is how the thesis reads it.
- **Why the reading inside a sealed laboratory is not absolute.** There are two cases. If no copy of the outcome leaves the laboratory, Wigner's record does not depend on that outcome, and its common part with the friend's reading is zero (in the model this is a check, true by construction). If instead Wigner undoes from outside the operation carried out in the laboratory, or performs an interference measurement on it as a whole, quantum mechanics does not even provide a single table of the probabilities of the two outcomes, and the common part is not defined: this is the same point at which, according to Cepollaro and Di Biagio, Aumann's theorem can fail. In both cases the thesis does not declare the friend's reading absolute: it is consistent with what the Wigner's-friend theorems call into question, the absoluteness of events that nobody compares.
- **Why in everyday life existence looks absolute, even though it is graded.** In the model, with a 1-bit object and a sharp tolerance, the common part goes from zero to the whole object within a few tens of copies, and ordinary objects leave hundreds of millions of copies per microsecond: collecting a tiny fraction of them is enough to be above threshold. With an object with more values and locally blurred copies, the common part computed within the class of readings tested grows in steps, but not according to the divisors of the grid: the case is calculated, in a classical model, in §9.
- **Why an object has a distance from me.** Distance is read by comparing two of my viewpoints on the same object (triangulation): what counts is the distance from the observer, and distances between objects follow from those. Triangulation is standard geometry and does not translate the "fractionation" the author speaks of; reading from the fraction of copies collected translates it more closely, but needs the total number of copies emitted by the source.

2.5 What it costs, said plainly

- There is no "object in itself" separate from all records. What has no non-local link with us is not a hidden object: for us it does not exist (in the model, the common part of the records is an indication of that link, not its definition). This is the side of the thesis that looks paradoxical, and it is what sets it apart.
- Existence for another becomes graded. In the 1-bit model the degree is almost always zero or full (§4), but in principle there are intermediate cases.
- It is not idealism: observers are not minds, records are physical states, the common part is a physical correlation between two records. And it is reciprocal: if A has a common part with the object, the object has the same common part with A.

2.6 What is new with respect to the closest studies

As far as we have found, no study defines the observed object itself as the common part of the records of two observers; the search covered arXiv abstracts and the texts cited here, and does not prove that none exists. The differences from the closest works:

- **Zurek** starts from a given system, with its pointer states (the states that withstand the interaction with the environment and that the environment copies), and from an environment that carries its copies; existence is the redundancy of the system's copies. Here no system is chosen in advance: the common part is defined from the records of any two parts (the chair itself is one of these parts), and it is symmetric, also between the object and the observer.
- **Rovelli, and Adlam and Rovelli,** relativise facts to systems and postulate the agreement between perspectives; here agreement becomes a quantity that is computed.
- **Leifer and Duarte** (2026), at the end of their paper and as an explicitly speculative remark, suggest that a shared objective reality emerges from common knowledge. This is the idea closest to the thesis: here it becomes a definition (the common part of the records) and is tested on a model.
- **Healey** (2013) objects to quantum Darwinism that consensus is not verification: many who agree do not make

true what they agree on. The thesis keeps apart the two things of §2.1: the common part between A and B (what they agree on) and the common part between the chair and each of them. They coincide when the two read independent copies above threshold; if they read the same defective copies, they would agree on something that is not what of the chair exists for each of them. That is the consensus without verification Healey speaks of, and the thesis recognises it as such.

- **Ontic structural realism** (Ladyman and Ross 2007) holds that relations come before objects. The thesis is a precise and measurable case of it: for two parts of the structure the object is a relation between their records, and its size is given in bits.
- **QBism** (Fuchs, Mermin and Schack 2014) makes an agent's probabilities and outcomes personal; here there are no agents and no personal probabilities.
- **Aumann and Contreras-Tejada et al.** speak of agreement between rational agents; the mathematical structure they use (the meet of the partitions) corresponds, on the finite space of outcomes, to the Gács–Körner common part. It is a correspondence of structure, not of concepts.

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3. How the thesis was tested

3.1 The model

The same kind of model as in quantum Darwinism: a quantity X of an object with two values (two positions), each with probability 1/2, that is, exactly 1 bit of information (a bit is the information of a choice between two equally likely possibilities); N carriers (scattered photons) each carrying a copy. Classical version: each copy is wrong with probability ε. Quantum version: the state $(|0\rangle|e_0\rangle^{\wedge}\otimes N + |1\rangle|e_1\rangle^{\wedge}\otimes N)/\sqrt{2}$, with overlap $s = \langle e_0|e_1\rangle$ (real, between 0 and

1. per carrier. In plain words: the two possible values of the object, 0 and 1, each leave their own imprint on N

photons, and the overlap s measures how similar the two imprints are, hence how hard they are to tell apart (s = 0: completely different imprints; s = 1: identical imprints). Two observers read m_A and m_B carriers; the reading is the best estimate (majority vote in the classical case; in the quantum case the Helstrom measurement, the best way allowed by quantum physics to tell two states apart). The dimension of space (three) is an input.

What the model uses and what the definition uses. The model starts from a source X and its carriers because this is how correlations between records arise in physics: through the light scattered by an object. The definition of the common object, instead, uses only the joint distribution of the two final records, P(R_A, R_B), that is, the table of the probabilities of all possible pairs of records: once the records are formed, it is computed without knowing anything about the source (criterion C4). The thesis does not deny the physical interactions that produce the records; it denies that an extra object is needed, beyond the parts of the structure and their common parts.

Three complementary mini-models. Two formulations and one check do not use photons. T-3 uses two grids of a and b intervals on the same segment, like two rulers with different divisions: the common part in its simplest, arithmetic form. T-7 uses the geometry of spheres and triangles in three-dimensional space, for distance. The check with moving observers uses events in the spacetime of special relativity.

3.2 The eight formulations

Each formulation translates into mathematics three of the author's words (they are in the background file): an observer's "matrix", which here, by a choice of the AI, is its raw record; the "projection", which here is the reading (§2.2); and the part "equal in large percentage", which here is the common part:

	formulation	in one line
T-0	no new consequence	quantum mechanics with cross-perspective links
T-1	the same matter	the common part is the set of carriers read by both
T-2	copies of the same fact	two copies of the same fact count as the same fact; each observer needs enough of them
T-3	the common divisor	each observer has a grid of a (or b) intervals; the common part is the common grid
T-4	the common part of	the Gács–Körner common part, exact (T-4s) or with tolerance δ (T-4δ); and a graded measure, the mutual information

	the records	between the readings (T-4g), which says how correlated the two readings are but is not itself a common variable
T-5	chains of copies	two observers have a common object if a chain of copies connects them
T-6	the same pattern	the common part is the similarity of the two distributions of readings
T-7	distance	from the fraction of copies collected (luminosity) or from comparing two viewpoints (triangulation)

3.3 The criteria, fixed in advance

C1, compared records agree (in version 2.62 the criterion had a second part added by the AI, removed in this version); C2, consistency with the position already published by the author on Wigner's-friend experiments (deposit 2.48: readings made in a laboratory from which no copy leaves are not absolute); C3, reciprocity; C4, no object given in advance (the common part is computed from the records alone); C5, the distance from the observer, with the dimension as an input; C6, declared parameters; C7, limiting cases (equal records: a single observer; no link, rendered in the model by two independent sources: no common object). Nine test cases, among them: different photons, sealed laboratory, comparison, two independent coins, distance, reciprocity, moving observers.

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4. Results

Three examples that can be done by hand.

- A ruler divided into 6 parts and one divided into 9 have in common the divisions of a ruler in 3 parts. The calculation confirms it for all 40,000 pairs of grids between 1 and 200: the common part of two grids of a and b intervals is the grid of gcd(a, b) intervals.
- Two people see the same chair with different photons: with disjoint carriers the overlap of the carriers is zero, yet the agreement is $1 - 8.6 \times 10^{-7}$ with 63 carriers each (s = 0.9). So the agreement does not come from shared matter. The reading "the same matter" (T-1) would say that the two have nothing in common. The author's criterion C1 concerns compared records only, so this datum does not exclude T-1; T-1 does not conform to criteria C4 and C5.
- Two coins give the same frequencies but are not the same coin: the similarity of the patterns is 1 even for two independent coins (T-6 fails). Having the same statistics is not having a common part: "the same pattern" does not distinguish two independent coins from two records of the same object.

In the 1-bit model, the common part with tolerance behaves like a yes-or-no. With copies that are wrong 10 % of the time:

tolerance δ	copies per observer above which the common part is essentially the whole object (≥ 0.9999998 bit)
10^{-3}	11
10^{-6}	23
10^{-9}	37

Below these thresholds the common part is zero (within the class of readings tested: readings that give 0 or 1 and depend only on how many copies say 0 and how many say 1). The thresholds depend on the quality of the copies: with poorly distinguishable quantum copies (overlap s = 0.99), an error of 10^{-6} needs 619 copies per observer; everyday objects leave hundreds of millions of copies per microsecond. The exact common part, without tolerance, is zero as soon as the copies are imperfect, even when agreement is very likely; with the comparison (B faithfully copies A's reading) it becomes 1 bit. Read this way (T-4s), the exact common part says that in the model, with imperfect copies, without comparison nothing is common in the strict sense and that the comparison makes it common: it is compatible with the data and with the position of deposit 2.48.

The "large common part". It is large if measured on the part of the record that concerns the object: at 23 copies the record carries 99.9998 % of the information about the object (0.9999975 bit out of 1), and the common part of the two records is about 1 bit (above threshold, at least 0.9999998 bit): in practice everything the record says about the object is common. On the whole record, which at 23 copies holds about 11.8 bits, the common part is small: 8.5 % at 23 copies and 3.4 % at 61: the rest of each record belongs to whoever holds it. This is the difference between the whole record and

its projection (§2.2). The whole record here is the raw record of the model, a choice of the AI, not the author's matrix. If two observers see the objects within a radius R and stand at distance D, the fraction of objects they have in common is 1 at D = 0, 0.46 at D = R/2, 0.19 at D = R and 0 at D = 2R (uniform density, sharp threshold).

The number of copies depends on their quality, not on how many there are. With N from 10^3 to 10^6 carriers, the flat stretch ("plateau") of the mutual information always starts at 65 copies (s = 0.9, with a deficit of 10^{-6} bit, that is, when less than a millionth of the information about the object is missing). This is the estimate, valid for many copies, of Zwolak, Riedel and Zurek (2014): ln(1/δ) divided by the quantum Chernoff information of one copy. For pure states the latter is exactly $-\ln|\langle e_0|e_1\rangle|^2 = -2 \ln s$, the rate at which the squared overlap decays with each extra copy; for s = 0.9 and $\delta = 10^{-6}$ the ratio gives about 66. With a deficit of 1 %, 21 copies are needed (s = 0.9), 213 (s = 0.99), 2140 (s = 0.999).

The formulations, one by one.

formulation	outcome
T-0	reference
T-1 the same matter	does not conform (C4, C5); in version 2.62 "excluded" by a part of C1 added by the AI
T-2 copies of the same fact	cannot be judged in the model: identifying the nexus with the source variable was a choice of the AI (in version 2.62: "not reciprocal and with the object given in advance")
T-3 the common divisor	compatible as an arithmetic rule of the common part: it is computed from the joint distribution of the two readings alone (test P1); the criterion on distances (C5) is met only together with triangulation, as for T-4 (in version 2.62: "does not conform")
T-4s exact common part	compatible: in the model, with imperfect copies, zero without comparison and 1 bit when B faithfully copies A's reading (in version 2.62 "excluded" by the part of C1 added by the AI)
T-4δ common part with tolerance	compatible, within the class of readings tested; the tolerance δ is a declared parameter (C6); provisional classification (model with copies wrong 10 % of the time, equal fragments)
T-4g graded measure (mutual information)	compatible, as a measure of how correlated the readings are
T-5 chains of copies	cannot be judged in the model, like T-2 (in the AI's rendering, with the nexus identified with the source variable, it would yield a common object even with a single noisy copy; the result does not judge the thesis)
T-6 the same pattern	excluded as a definition of the object: it does not distinguish two independent coins from two records of the same object
T-7L distance from luminosity	distance only, if the total number of copies emitted is known (a datum distinct from the mass, which in the model is not derived from it); criterion C3 does not apply to a distance
T-7P distance by triangulation	gives the distance, given the baseline and the angles: standard geometry, which does not translate the "fractionation"

Moving observers (check only). What all inertial observers have in common is the causal order of events; the simultaneity of distant events is not common. This is relativity: the calculation recovers it on a sample (at v = 0.6c, the same order for 83 % of the pairs, timelike pairs 50 %), and the general result is analytic.

Predictions that did not succeed, declared. We had predicted that, with copies that are wrong 10 % of the time, 21 copies would suffice at tolerance 10^{-6}: 23 are needed. We had predicted intermediate values of the common part below threshold: within the class of readings tested, with a 1-bit object, there are none. With an object with more values, within the class of readings tested, there are: they are the steps of §9. A numerical error in the first program (a subtraction of nearly equal numbers) was corrected in a separate file; the other calculations do not change.

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5. What is not claimed

- Objectivity is not derived from the premises, and the measurement problem is not solved.
- The probabilistic definitions tested are functions of the distribution of the records given by quantum mechanics:

in the cases examined, by construction, they give no different statistics; the common divisor and triangulation are arithmetic and geometric rules, not statistical ones. The thesis is an interpretation consistent with the data, not yet a test.

- The link with spectrum broadcast structures is only this: for classical records perfectly distinguishable in a fixed basis, the branch label is a common variable of every pair of records. Spectrum broadcast structures require more.
- The correspondence with Aumann is one of mathematical structure, not of concepts.
- The non-local phase is the author's interpretive framework: the calculations of this work do not test it. They show that the reading "the common object is the common part of the projections" is consistent with the data; that underneath there is a non-local relation, and not only the physics of scattered light, is an interpretation.
- It is not claimed that the chair disappears from the model. The source and its carriers serve to produce the distribution of the records, and the distinction between seeing and consensus without verification (Healey) uses the common part between the chair and each observer. The thesis treats the chair as one of the parts of the structure, with a state of its own; what it removes is an additional common object, given before the records.
- The photon model translates the projection; the author's matrix of multiples and fractions, the nexus and the fractionation have no full counterpart in it. This is why copies of the same fact and chains of copies cannot be judged in the model, and why the common divisor and triangulation remain two separate mini-models, one arithmetic and one geometric (in the classical model of §9 the common divisor becomes the ceiling or a stage of the common part).
- Not studied here: records with few noisy copies in models other than this one; partial leakage of copies in a Wigner scenario (in quantum mechanics it was studied by Rivlin, Engineer and Baumann 2025, but what the thesis would say there has not been calculated); larger friends. There the thesis would give a criterion of existence, not yet a different statistics.

6. The contact with mass

The author's framework also ties mass to the multiple, that is, to counting in a row; that link is treated in a separate work (2.64), and that counting is not the number of photons emitted by a source. In this model the definition of the common object does not use mass, because the copies counted are taken at the same moment, side by side: this is a property of the static model. Real copies accumulate in time (§1), that is, "in a row": whether the projection can be defined without the multiple remains open. Distance from luminosity needs the total number of copies emitted, a datum distinct from the mass, which in the model is not derived from it (in the case considered it depends on the illumination and the temperature of the source).

7. How to reproduce

Four Python 3 programs (standard library and numpy), under a minute in total on an ordinary computer: livello1_conti.py (the registered run), correzione1_conti.py, correzione2_conti.py, correzione3_conti.py. Every protocol and every program has a SHA-256 fingerprint and a UTC time registered before the run; the results are in JSON files. The protocols, the corrections and the two independent verifications are in the package. Section 9 needs three more Python 3 programs (with numpy), about eighty minutes in all: oggetto_piu_valori_conti.py, correzione1_conti.py and correzione2_conti.py, in the folder 05_oggetto_piu_valori with the protocol, the corrections, the outcome notes and the two independent verifications. In the working records the files keep their Italian names.

8. Provenance

The author is not a physicist. The formulations, programs and calculations were made with an artificial intelligence (Claude, Anthropic) and checked by a second one (gpt-6-luna via Codex, OpenAI), in two rounds; of the 25 points raised in the two verifications, 24 were accepted and one was rejected with an exact calculation, then withdrawn by the verifier. The predictions were registered before the calculations. The author's sentences from which the work arises are in the background file. In version 2.62.1 an independent check requested by the author (another instance of Claude, which did not know the conclusions of the first) found the undeclared translation choices corrected in this version; each point was rechecked against the original files. The corrections were then verified by the second AI; the reports and the answers are in 04_controllo_262_264. An external review of the corrected version then added two clarifications to §5. Section 9, suggested by the same review, was verified by the second AI in two rounds.

9. An object with more values

New section in version 2.62.1. It answers a limit pointed out by an external review: in the model of the previous sections the object is worth 1 bit, and the common part goes from zero to the whole object all at once.

What was tested. A classical model, like the classical version of §3.1: the object is a position on a grid of n cells (6 or 12), and each of the two observers collects m noisy, independent copies of it. There are two questions: does the common part with tolerance arise in steps, coarse fractions first and fine ones later? And do the steps follow the divisors of the grid, like the rulers of the common divisor (T-3)? The common part is computed among readings that divide the grid into blocks (a **partition**) and answer with the most probable block; with 12 cells all partitions into contiguous intervals are tried. This is the **class of readings tested**: it gives a lower bound on the common part with tolerance, because readings of another kind might find more. The noise has a single value for each type: copies blurred by about one cell (a bell curve one cell wide), or copies that err by one step towards a neighbouring cell, to the right or to the left, 20 % of the time, or copies that err towards any cell. The translations are choices of the AI: the author's «fractions» are rendered by the partitions of the grid, and the «coming into being» of the fractionation by the growth of the common part as the copies increase. Protocol, expectations and programs were registered before the calculations, and the second AI verified them in two rounds (folder 05_oggetto_piu_valori). With a 1-bit object the model recovers the thresholds of §4: 11, 23 and 37 copies for $\delta = 10^{-3}$, 10^{-6} and 10^{-9}.

Prior work. The common part is that of Gács and Körner (1973), defined for arbitrary discrete variables. Witsenhausen (1975) shows that the exact common part vanishes when the joint distribution has a connected support, as in the noisy cases studied here. In discrimination among several hypotheses, with many copies the error exponent is fixed by the least distinguishable pair of hypotheses, the one with the smallest Chernoff information (Leang and Johnson 1997). Dithering in converters (Gray and Stockham 1993) is an analogy, not an equivalence, for the case of noise before the ruler (point 4). It is not claimed that an object with more values and the common part with tolerance has never been studied: a systematic literature search was not carried out.

1. Within the class tested the common part grows in steps. A grid of 12 cells, copies blurred by about one cell, allowed disagreement $\delta = 10^{-6}$:

- up to 81 copies per observer, no common part within the class tested;
- from 82 copies, the halves (1 bit); from 88, the thirds (1.585 bits); from 91, the quarters (2 bits);
- then 5, 6, 7… parts, up to the whole object (3.585 bits) at 101 copies.

One step for each additional boundary: eleven steps between 82 and 101 copies. With $\delta = 10^{-3}$ the ladder runs from 31 to 48 copies, with 10^{-9} from 136 to 155: the more agreement is required, the narrower the ladder becomes in proportion. With copies that err by one step 20 % of the time the ladder fits within 2-4 copies (from 19 to 23 with $\delta = 10^{-6}$). For ordinary objects, which leave hundreds of millions of copies, this is an estimate and not a calculation: if the copies stay independent and the noise and the tolerance stay fixed, the ladder is in practice a yes-or-no, as in §4.

2. The steps do not follow the divisors. At each step a boundary between cells is added, and the cost of a boundary (the probability that the two readings end up on its two sides) is almost the same wherever it lies; the complete enumeration of the class tested gives as best the division into parts as equal as possible. The parts are equal when their number divides 12 (halves, thirds, quarters, sixths, twelfths); with 5, 7, 8, 9, 10 and 11 parts they are unequal (for example 3 + 3 + 2 + 2 + 2), and the steps occur all the same. The divisors give the «clean» steps, not the rule. Moreover, the new parts do not always lie inside the old ones: from the halves one passes to the thirds, which cut across them.

3. «Coarse fractions first» comes from the blurring. If the copies err towards any cell instead of towards the neighbouring ones (grid of 6 cells), halves and thirds are never the best common part: the first step isolates one cell or divides into unequal parts. Coarse fractions come first only with local noise, because nearby values are confused more than distant ones.

4. The common divisor is the ceiling or a stage, depending on where the noise enters. Two observers with different rulers, of a and b marks on the same grid (6 and 9, or 12 and 18), with the same noise (one step, 20 % of the time):

- if the noise enters after the ruler (each copy carries the mark, sometimes wrong by one), the common part grows up to the common ruler of GCD(a, b) marks and stops there: 1.585 bits for 6 and 9, 2.585 bits for 12 and 18. Within the class of readings tested it is an exact ceiling. For every possible reading, in the model studied (equally probable positions, up to 60 copies), an analytic bound exceeds it by at most 0.0011 bits with $\delta = 10^{-6}$; with $\delta = 10^{-3}$ the bound is weak and not enough to say «ceiling». This is the common divisor of §3 (T-3) as the ceiling of the common part with tolerance;
- if the noise enters before (each copy carries the position blurred by one fine cell, then read on the ruler), with $\delta = 10^{-6}$ the common part reaches the common ruler at 21 copies, stays there for a long stretch and then goes beyond it: A's ruler at 115 copies, B's at 121, the fine grid at 133. Copies that spill into the neighbouring mark let each observer read finer than their own marks. Here the common divisor is a stage, not the ceiling.

5. The other measures. The exact common part, without tolerance, is zero with noise, as in §4 (checked with 1 and 2 copies in all cases). The graded measure (the mutual information between the two readings) grows smoothly, without

steps: with 12 cells it is 2.680 bits at 10 copies and 3.5847 bits at 80, when the common part with $\delta = 10^{-6}$ is still zero within the class tested. The steps are a property of the requirement to agree almost always.

What it says for the thesis. The reading «the common part arises in steps, coarse fractions first» is compatible in the model, within the class of readings tested and with locally blurred copies. The reading «the steps are the divisors» does not follow: this reading of the fractionation falls, not the thesis of the common part. The common divisor of the two GCDs joins the common part of the copies as a ceiling if the noise enters after the ruler, as a stage if it enters before; which of the two cases translates the author's thought is a choice, not decided here.

Limits. The model is classical; the class of readings is restricted, so the numbers are a lower bound on the common part (only with the noise after the ruler is there also an upper bound); each noise parameter has a single value, and the numbers of copies hold for those values.

Glossary

- **Absoluteness of Observed Events:** the assumption that what an observer has seen is a single fact, the same for everyone.
- **Bit:** the information of a choice between two equally likely possibilities. An object that can be in two places with equal probability carries 1 bit.
- **Causal order:** which event can influence which; for pairs of events that can influence each other it is the same for all moving observers, unlike the simultaneity of distant events.
- **Common knowledge:** each knows something, knows that the other knows it, knows that the other knows that they know it, and so on.
- **Common part (Gács–Körner):** the richest quantity that each of the two can compute from its own record, with the same value as the other. With tolerance δ: the same value except with probability δ.
- **Copy, carrier:** a piece of the environment (here a scattered photon) that carries an imprint of the object's value.
- **Cross-perspective links:** the postulate of Adlam and Rovelli: if B measures the variable in which A recorded an outcome, B finds A's outcome.
- **Decoherence:** the disappearance of interference in the behaviour of the system alone when it interacts with its environment; coherence is not destroyed, it passes into the correlations with the environment.
- **Helstrom measurement:** the best way allowed by quantum physics to tell two states apart.
- **Joint distribution:** the table of the probabilities of all possible combinations of two (or more) records.
- **Local Friendliness:** a set of assumptions about Wigner's-friend scenarios (absoluteness of observed events, no superdeterminism, locality) and the bounds they imply for the measured correlations.
- **Meet of partitions:** among the subdivisions that both can recognise with their own information (that is, coarser than each one's), the finest.
- **Mutual information:** how much information one record gives about another (or about the object), in bits.
- **Non-local phase:** in the author's framework, the level of relations without space and without time, present in every place and at every instant; "non-local" means "not located in space", not "influence at a distance". The local world is the same reality counted from a point.
- **Observation matrix:** the author's word; in the author's framework multiples and fractions of the common unit form a matrix, and here it is read as the structure of counting seen from an observer's point of view (a reading of the AI). In the model of this work it is represented, by a choice of the AI, by the raw record of the photons collected.
- **Observer:** any physical system that records and compares; not a mind.
- **Ontic structural realism:** the philosophical thesis that relations come before objects.
- **Overlap s:** how similar the imprints left by the two values of the object are (0: completely different; 1: identical).
- **Partition:** a division of the grid into blocks; in §9, a reading that says in which block the object lies.
- **Plateau:** the flat stretch of a curve; here, the point beyond which reading more copies adds nothing about the object.
- **Pointer states:** the states of a system that withstand the interaction with the environment and that the environment copies.
- **Projection:** the passage from something rich to something poorer that keeps only some of its features, like the shadow of a body on a wall; here, the reading an observer makes of its own record.
- **QBism:** an interpretation of quantum mechanics in which probabilities and outcomes are personal to the agent who uses them.
- **Quantum Chernoff information:** how much a single copy helps to tell the two possible values apart; for pure copies with overlap s it equals $-2 \ln s$.
- **Quantum Darwinism:** the theory (Zurek) according to which a fact is objective when the environment carries many independent copies of it, which many observers can read without disturbing the object.

- **Reading:** what an observer extracts from its record: a projection of the record.
- **Record:** everything an observer has recorded, as a physical state of its memory.
- **Redundancy:** how many independent copies of the same fact are in the environment.
- **Spectrum broadcast structure:** the state in which every fragment of the environment, on its own, carries the value of the system.
- **Tolerance δ:** the error one accepts; for example $\delta = 10^{-6}$ means one disagreement in a million cases.
- **Transposition:** in the author's framework, the fact that a part of one's own observation matrix presents itself as an object located elsewhere, at a distance; in the model the distance is read by triangulation.
- **Triangulation:** reading the distance of an object by comparing the directions in which it is seen from two points.
- **Wigner's friend:** a scenario in which one observer (the friend) measures something shut inside an isolated laboratory, while a second observer outside (Wigner, the "superobserver") treats the whole laboratory as a single quantum system and can even undo from outside what happened inside.

Package contents

- README.md (this text); LEGGIMI_IT.md (the same text in Italian); AUTHOR_BACKGROUND.md (the author's sentences, kept apart from the main text).
- 01_oggetti/LIVELLO1_PROTOCOLLO_v1.md: the eight formulations, nine test cases, criteria and predictions, registered before any calculation (in Italian, as registered).
- 01_oggetti/CORREZIONE_1_LIVELLO1.md, CORREZIONE_2_LIVELLO1.md, CORREZIONE_3_LIVELLO1.md: corrections, each registered before its program; NOTA_ESITO_LIVELLO1_v1.md, v2, v3: the outcome notes (v3 is the final one).
- 01_oggetti/programmi: the four programs; 01_oggetti/risultati: their JSON results and logs.
- 01_oggetti/VERIFICA_ASTRA_G1 and VERIFICA_ASTRA_G2: the two verification reports (gpt-6-luna via Codex).
- REGISTRO.txt in the main folder and in each subfolder: SHA-256 fingerprint and UTC time of every registration. A few registered working files are not in the archive because they quote the author's working sentences (the premises, the Level-0 inventories of sources, the verification prompts, which contain the whole material); their fingerprints are in the REGISTRO.txt files, and they can be shown on request. The sources are listed in the References; third-party papers are not redistributed.
- 05_oggetto_piu_valori (section 9): PROTOCOLLO_OGGETTO_PIU_VALORI_v1.md, the protocol registered before the calculations; CORREZIONE_1.md, CORREZIONE_2.md and CORREZIONE_3.md, the corrections and the answers to the verifications; NOTA_ESITO_v1.md to v3, the outcome notes (v3 is the final one); VERIFICA_ASTRA_G1 and VERIFICA_ASTRA_G2, the two verification reports; programmi, the three programs; risultati, their JSON results (all in Italian, as registered).
- 04_controllo_262_264/PIANO_CORREZIONI_v1.md: the plan of the corrections of version 2.62.1, registered before the texts were changed (in Italian); 04_controllo_262_264/REGISTRO.txt: fingerprints of the independent check and of its verification, which stay outside the archive because they quote the author's working sentences.
- 04_controllo_262_264/VERIFICA_ASTRA_262_1_G1 and following rounds: the reports of the second AI on the corrections of version 2.62.1 and the answers (in Italian).
- run_all.py: re-runs every calculation and compares the results with the registered ones; IMPRONTE.sha256: fingerprints of all the other files in the archive.
- LICENSE.md: texts under CC BY 4.0, code under the MIT licence.
- In the working records this deposit is called «oggetti osservati, Livello 1» (folder MM_sviluppo_INDIGERIBILI/01_oggetti); file names and registered documents keep their Italian names and language.

How to run

python run_all.py

The script runs the four programs of 01_oggetti and the three programs of 05_oggetto_piu_valori (section 9; about eighty minutes) into separate subfolders of a temporary folder and compares every number with the registered JSON files (relative tolerance 10^{-9} of the larger value, with no absolute floor). It ends with «ALL MATCH». Tested with Python 3.13 and numpy 2.4; about eighty minutes in all, almost all for section 9. A few registered numbers are not compared: they are listed in the script (computing times, which change at every run).

Related records

- MM 2.48, DOI 10.5281/zenodo.23352814: the author's placement with respect to Local Friendliness (criterion

C2).

- Status report, concept DOI 10.5281/zenodo.21802475, addendum of 26 August 2026 (QUANTUM_PLACEMENT.md, §0): the earlier reading of agreement as copies above a threshold, which this deposit develops.
- Entry page of the framework, DOI 10.5281/zenodo.22667516: postulate 2.

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